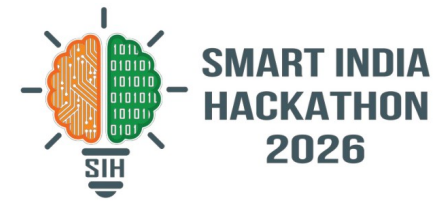
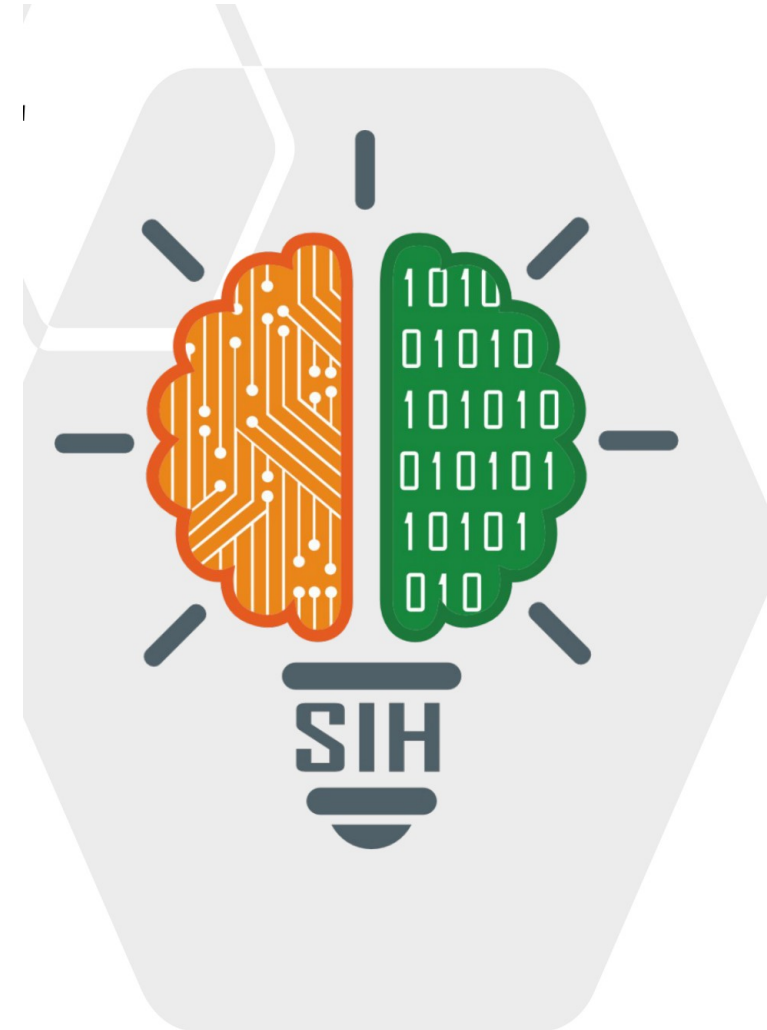


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – SIH26236**
- **Problem Statement Title – AI-Based Intelligent Food Packaging Material Recommendation System for Food Commodities**
- **Theme – Agriculture, FoodTech & Rural Development**
- **PS Category – Software**
- **Team ID – 161280**
- **Team Name (Registered on portal) – Evergreen**



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

PACKWISE AI — Physics-informed hybrid AI platform matching 100+ Indian food commodities & storage physics with 31 barrier polymers. ✓ Explainable • ✓ Source-traced • ✓ Respiration & MAP aware

100+ Indian Commodities DB (FDC+ICAR)

Mass-Transfer & Respiration Physics

Dynamic Storage & Transit Logistics

31 Polymers & Bio-Resins Catalog

TOPSIS Multi-Criteria Ranking

Dynamic Barrier Needs (OTR / WVTR)

MAP Equilibrium Solver (O₂ / CO₂)

Explainable Report, FSSAI Pass & QR

❖ How it addresses the problem

- **Perishable crop devastation** — India suffers ~30% post-harvest spoilage (₹1.52 Lakh Cr loss) due to unvented polybags and condensation rot.
- **One pack can't fit all** — Moisture sorption, fat oxidation, and respiration rates differ fundamentally across commodities; mismatched films cause premature spoilage.
- **Selection is expert-dependent & slow** — Fragmented offline knowledge, arbitrary polymer thickness, and high trial-and-error laboratory costs.
- **Ground-level stakeholders struggle** — Smallholder farmers, FPOs & food MSMEs lack packaging scientists and need instant, explainable guidance in their native language.
- **Unified physics + regulatory decision core** — Integrates food composition, crop biology, polymer barrier physics, and FSSAI/BIS rules into one automated engine.

SOLUTION PIPELINE — FROM FOOD TO EXPLAINED RECOMMENDATION

FOOD SELECTION
(100+ Crops)

PROPERTY ANALYSIS
(aw, pH, Lipids)

RISK ANALYSIS
(Respiration & O₂)

COMPATIBILITY
(31 Polymers)

TOPSIS AI RANKING
(Cost vs Eco)

EXPLAINED RESULT
(3D Lab + QR)

❖ Innovation and uniqueness of the solution

1. EXPLAINABLE AI

Transparent mathematical justification (TOPSIS Closeness C_i^*) with statutory FSSAI citations & Gemini Flash reasoning.

2. MASS-TRANSFER PHYSICS

Tetens vapor pressure equation + Arrhenius respiration kinetics for anaerobic hazard alerts.

3. 3D PACKAGING LAB

Interactive Three.js WebGL multi-layer film digital twin with real-time layer exploding + 50-frame storyboard.

4. SOURCE TRACEABILITY

Strict data discipline: FSSAI, APEDA, BIS, ICAR & USDA FDC — zero fabricated synthetic values.

5. AASAAN VOICE + QR

Bilingual Hindi/English voice assistant (Vaani AI) + Edge YOLOv8 & NVIDIA NIM camera crop scanner + instant batch QR pass.

❖ Technologies, AI Software & Models Used in the Platform

FRONTEND TECH STACK

- **Languages:** HTML5, CSS3, JavaScript (ES6+), TypeScript
- **Framework:** React 19, Vite (Instant HMR), Tailwind CSS
- **3D Digital Twin:** Three.js WebGL (Interactive Film Lab) & Quantum 3D
- **Sensory UI:** Web Speech API (Voice AI), WebRTC Camera Scanner

BACKEND & DATABASE

- **Core Runtime:** Python 3.12 (High-Performance Async)
- **API Framework:** FastAPI (RESTful APIs, Auto-Docs, CORS)
- **Server & Engine:** Uvicorn ASGI, Pydantic v2, SQLAlchemy
- **Database:** SQLite (Offline Local) / PostgreSQL Cloud

AI & MACHINE LEARNING

- **YOLOv8n Edge ONNX:** 12ms / 60 FPS real-time crop camera scanner
- **NVIDIA NIM Llama-3.2:** GPU Vision & packaging defect detection
- **Google Gemini Flash:** FSSAI statutory explainability & multimodal
- **NumPy TOPSIS:** MCDM Multi-Criteria Decision (Cost vs Eco)

PHYSICS & KINETIC MODELS

- **Tetens Equation:** $P_{sat}(T)$ Saturated Vapor Pressure & WVTR
- **Michaelis-Menten:** Dynamic Crop O₂ & CO₂ Respiration Rates
- **Arrhenius Model:** Temperature Kinetics & +5°C Safety Buffer
- **MAP Solver:** Laser Micro-Perforation Permeability

❖ System Architecture Flow — End-to-End Multi-Tier Connection

1. CLIENT / USER LAYER

- React 19 Responsive UI & Tailwind
- Bilingual Voice AI (Hindi / English)
- WebRTC Camera Crop Scanner

2. API & GATEWAY LAYER

- FastAPI Asynchronous Router
- Pydantic v2 Schema Validation
- REST Endpoints (/optimize, /recommend)

3. AI & PHYSICS PIPELINE

- YOLOv8 Edge ONNX + NVIDIA NIM
- Tetens & Arrhenius Kinetics Solver
- Gemini Flash Statutory Explanations

4. DATA & STANDARDS LAYER

- 100 Indian Crops (ICAR / USDA FDC)
- 31 Polymers & Bio-Plastics DB
- FSSAI 2018 (IS 9845) & BIS Standards

5. OUTPUT & DELIVERY HUD

- Interactive 3D Three.js Film Lab
- MAP Gas Balance (5% O₂, 10% CO₂)
- Digital Traceability Batch QR Pass

OPERATIONAL METHODOLOGY PIPELINE

1. **Crop Intake & Profiling:** Input crop & shelf life via Hindi Voice/Camera; extracts moisture %, water activity (aw), pH, lipids & baseline respiration rate.
2. **Physics & Respiration Modeling:** Tetens equation calculates vapor pressure $P_{sat}(T)$; Michaelis-Menten & Arrhenius model O₂/CO₂ rates with a +5°C transit safety buffer.
3. **Barrier Solving & Screening:** Computes target OTR (cc/m²·day) and WVTR (g/m²·day) bounds; filters 31 food-grade barrier polymers & bio-resins.
4. **TOPSIS MCDM Optimization:** Vector-normalized ranking balances Shelf-Life Extension (Max), Film Cost (Min), and Carbon Footprint (Min).
5. **Regulatory Audit & 3D Deliverable:** Verifies FSSAI 2018 (IS 9845) migration compliance; renders Three.js 3D multi-layer pack and batch QR pass.

INTELLIGENT MULTI-LAYER FOOD PACKAGING ARCHITECTURE

Outer Protective Layer: BOPET (12 μm) → Mechanical puncture resistance & printable surface

Adhesive Tie Layer: Modified PE (3 μm) → Inter-layer molecular bonding preventing delamination

High Gas Barrier: EVOH Core (5 μm) → Ultra-low OTR (<1 cc/m²·day) • Prevents oxidation & rancidity

Adhesive Tie Layer: Modified PE (3 μm) → Inter-layer molecular bonding preventing delamination

Hermetic Sealant: Metallocene LLDPE (50 μm) → Moisture barrier (WVTR <3 g/m²·day) • FSSAI Food Contact Safe (IS 10146)

MAP Gas Equilibrium: Laser micro-perforations match respiration (RO₂) with film gas flux (3-5% O₂, 10-15% CO₂) to extend produce shelf life by 200% - 300%.

❖ Analysis of the feasibility of the idea

TECHNICAL FEASIBILITY

- ✓ Modern, battle-tested full stack — React 19 + FastAPI
- ✓ Sub-50ms execution latency for TOPSIS & physics algorithms
- ✓ Modular, high-speed RESTful microservices architecture
- ✓ Three.js WebGL runs smoothly in any standard mobile/PC browser
 - ▶ **100% functional working prototype verified locally & container-ready.**

DATA RIGOR & PROVENANCE

- ✓ 100 pre-calibrated commodities across 6 core food categories
- ✓ Integrated official FSSAI • APEDA • BIS • ICAR-CIPHET standards
- ✓ Complete provenance lineage kept for every single value & equation
- ✓ Strict zero synthetic/hallucinated data — fully explainable calculations
 - ▶ **Every rule, threshold & barrier value traceable to Slide-6 sources.**

DEPLOYMENT & SCALING

- ✓ Fully functional cloud & edge deployable prototype with offline mode
- ✓ Zero proprietary hardware Capex needed (any smartphone or PC)
- ✓ Stateless FastAPI containers scale horizontally with traffic load
- ✓ Direct integration pathway with e-NAM digital mandis & APEDA
 - ▶ **Built for immediate grassroots deployment across rural FPOs.**

❖ Potential challenges and risks → Strategies for overcoming these challenges

1. Crop respiration & moisture vary by harvest maturity → Calibrated against ICAR & UC Davis postharvest baselines; live sliders allow fine-tuning.

2. Broken Indian rural cold-chain transit abuse → Arrhenius kinetic engine factors in +5°C transit safety buffer to prevent premature spoilage.

3. Material barrier variations across film suppliers → Normalized ASTM/ISO testing metrics (OTR: ASTM D3985, WVTR: ASTM F1249) stored in DB.

4. Complex Plastic Waste Management (PWM) Rules 2022 → Automated regulatory audit checks FSSAI (IS 9845) migration and prioritizes compostable PLA/PHA.

5. Rural digital literacy divide for grassroots farmers → Aasaan Mode with native Hindi voice assistance and real-time camera crop recognition.

✓ **Verdict: Fully feasible with modern web & physics AI stack — validated working prototype ready for immediate pilot. • Risk: Low–Medium**

❖ Potential impact on the target audience

Farmers / Mandi
FPOs

Food Processors

Agri-MSMEs

Packaging Designers

Exporters

Cold-Chain Logistics

Food Scientists

Govt + Extension

❖ Benefits of the solution (social, economic, environmental, etc.)

SOCIAL

- ▶ Democratizes packaging science for grassroots farmers
- ▶ Native Hindi/English voice AI bridges literacy barrier
- ▶ Prevents distress sales during market glut via shelf-life extension
- ▶ Protects smallholder farm incomes & rural livelihoods

ECONOMIC

- ▶ Reduces post-harvest spoilage by 25% – 40% across transit
- ▶ Mitigates India's annual ₹1.52 Lakh Crore food loss burden
- ▶ Saves 12-20% resin cost via precision film down-gauging
- ▶ Eliminates costly export consignment rejections at ports

AGRICULTURAL

- ▶ Crop-wise preservation for 100 Indian commodities
- ▶ Preserves fresh produce vitamins, firmness & natural aroma
- ▶ Enables long-distance rail/road freight without spoilage
- ▶ Generates export-ready APEDA packaging compliance passes

ENVIRONMENT

- ▶ Recommends compostable Bio-PLA, Bio-PBS & PHA films
- ▶ Accelerates transition to recyclable mono-material polyolefins
- ▶ Fully aligns with Plastic Waste Management (PWM) Rules 2022
- ▶ Minimizes methane emissions from rotting mandi landfill waste

FOOD QUALITY

- ▶ Optimal OTR/WVTR stops moisture pooling & fungal rot
- ▶ MAP gas equilibrium (O₂/CO₂) preserves crispness & color
- ▶ Prevents lipid oxidation and rancidity in high-fat foods
- ▶ Guarantees certified, safe & hygienic produce for consumers

ADOPTION PATHWAY — PROTOTYPE TO SCALE

1. PROTOTYPE (Functional Full-Stack)

2. PILOT (Selected Mandis & FPOs)

3. VALIDATION (IIP & Lab Trials)

4. SCALE-UP (National e-NAM / Cloud)

Compare Recyclable & Compostable Options

Right-Size & Down-Gauge Packs (Less Plastic)

Voice-First, Low-Literacy Bilingual UI

Rural + Mandi FPO Friendly Practical Guidance

❖ Details / Links of the Reference, Regulatory & Research Work

FUTURE SCOPE & ADVANCED EXTENSIONS

- 1. National e-NAM Mandi Integration:** Direct API integration with electronic mandis for automated packaging grading and instant transit passes.
- 2. IoT Telemetry & Cold-Chain Sensors:** Live RFID/BLE temperature & RH data streams for real-time dynamic shelf-life decay updates.
- 3. Active & Antimicrobial Bio-Films:** Integration of natural nano-emulsion coatings (chitosan, thymol) to actively arrest fungal decay.
- 4. Converter Marketplace & Procurement:** Direct B2B digital gateway connecting FPOs directly with verified sustainable film converters.
- 5. Offline Progressive Web App (PWA):** Zero-connectivity Service Worker caching for seamless use in remote rural mandis.
- 6. AI Computer Vision Ripeness Grading:** Camera-based maturity & blemish detection prior to barrier film recommendation.

OFFICIAL GOVERNMENT & SCIENTIFIC SOURCES (VERIFIED PORTALS)

- FSSAI — [\[https://fssai.gov.in\]](https://fssai.gov.in) : Packaging Reg. 2018 & Overall Migration Limits (IS 9845:2020)
- BIS — [\[https://bis.gov.in\]](https://bis.gov.in) : Indian Standards: IS 10146 (PE food contact), IS 12252 (PET), IS 15495
- APEDA — [\[https://apeda.gov.in\]](https://apeda.gov.in) : Fresh produce export packaging standards & cold-chain phytosanitary norms
- IIP — [\[https://iip-in.com\]](https://iip-in.com) : Indian Institute of Packaging (ASTM D3985 OTR & ASTM F1249 WVTR)
- ICAR - CIPHET — [\[https://ciphnet.icar.gov.in\]](https://ciphnet.icar.gov.in) : Post-harvest loss statistics & crop respiration rates (RO₂/RCO₂)
- MoFPI — [\[https://mofpi.gov.in\]](https://mofpi.gov.in) : Ministry of Food Processing Industries cold-chain cluster schemes
- USDA FoodData — [\[https://fdc.nal.usda.gov\]](https://fdc.nal.usda.gov) : Crop baseline composition dataset (moisture %, Aw, lipids, pH)
- CPCB (MoEFCC) — [\[https://cpcb.nic.in\]](https://cpcb.nic.in) : Plastic Waste Management Rules 2022 & Compostable IS/ISO 17088

EVIDENCE DISCIPLINE — ZERO FABRICATED SYNTHETIC VALUES

OFFICIAL / VERIFIED

AUTHORITATIVE RESEARCH

MANUFACTURER SPEC

DERIVED PHYSICS

NOT AVAILABLE (FLAGGED)

PROTOTYPE DEMO PLAN — WHAT WE WILL DEMONSTRATE IN EVALUATION

1. Live 100-Crop Search & Camera Scan

2. Mass-Transfer & Respiration Solver

3. TOPSIS Dynamic Optimization & 3D Lab

4. Bilingual Voice AI & FSSAI QR Pass



PROTOTYPE LINK — <https://food-packaging-ai-caed.vercel.app/>